

LAB02.JoséMario

Questão 1)

Precisamos guardar duas estatísticas suficientes, a soma dos voos atrasados, e o total de voos contados, então ai sim somamos todos os atrasos, os contados, ai calculamos a porcentagem final.

```
#Questão 2)
getStats <- function(input,pos) {
  input %>% filter(AIRLINE %in% c('AA', 'DL', 'UA', 'US')) %>%
  filter(!is.na(ARRIVAL_DELAY), !is.na(YEAR), !is.na(MONTH), !is.na(DAY), !is.na(AIRLINE)) %>%
  group_by(YEAR, MONTH, DAY, AIRLINE) %>%
  summarise(atrasos = sum(ARRIVAL_DELAY > 10), total = n(), .groups = "drop")
}
```

```
# Questão 3)
colunas_interesse <- cols_only(
  YEAR = col_integer(),
  MONTH = col_integer(),
  DAY = col_integer(),
  AIRLINE = col_character(),
  ARRIVAL_DELAY = col_double()
)
dados_resumidos_lotes <- read_csv_chunked(
  file = "C:\\Users\\José\\Documents\\ME315\\Nova pasta\\flights.csv",
  callback = DataFrameCallback$new(getStats),
  chunk_size = 100000,
  col_types = colunas_interesse
)
head(dados_resumidos_lotes)
```

```
# A tibble: 6 x 6
  YEAR MONTH DAY AIRLINE atrasos total
  <int> <int> <int> <chr>    <int> <int>
1  2015     1     1 AA         514  1379
2  2015     1     1 DL         132  1558
3  2015     1     1 UA         367  1269
4  2015     1     1 US         201  1028
5  2015     1     2 AA         649  1508
6  2015     1     2 DL         406  2261
```

```
# Questão 4)
computeStats <- function(dados) {
  dados %>%
    group_by(AIRLINE, YEAR, MONTH, DAY) %>%
    summarise(
      atrasos_totais = sum(atrasos, na.rm = TRUE),
      voos_totais = sum(total, na.rm = TRUE),
      .groups = "drop"
    ) %>%
    transmute(
      Cia = AIRLINE,
      Data = as.Date(paste(YEAR, MONTH, DAY, sep = "-")),
      Perc = atrasos_totais / voos_totais
    )
}
stats <- computeStats(dados_resumidos_lotes)
head(stats)
```

```
# A tibble: 6 x 3
  Cia Data      Perc
  <chr> <date>    <dbl>
1 AA  2015-01-01 0.373
2 AA  2015-01-02 0.430
3 AA  2015-01-03 0.538
4 AA  2015-01-04 0.602
5 AA  2015-01-05 0.491
6 AA  2015-01-06 0.366
```

```
#Questão 5)
library(ggcal)
library(ggplot2)
```

```

pal <- scale_fill_gradient(low = "#4575b4", high = "#d73027")

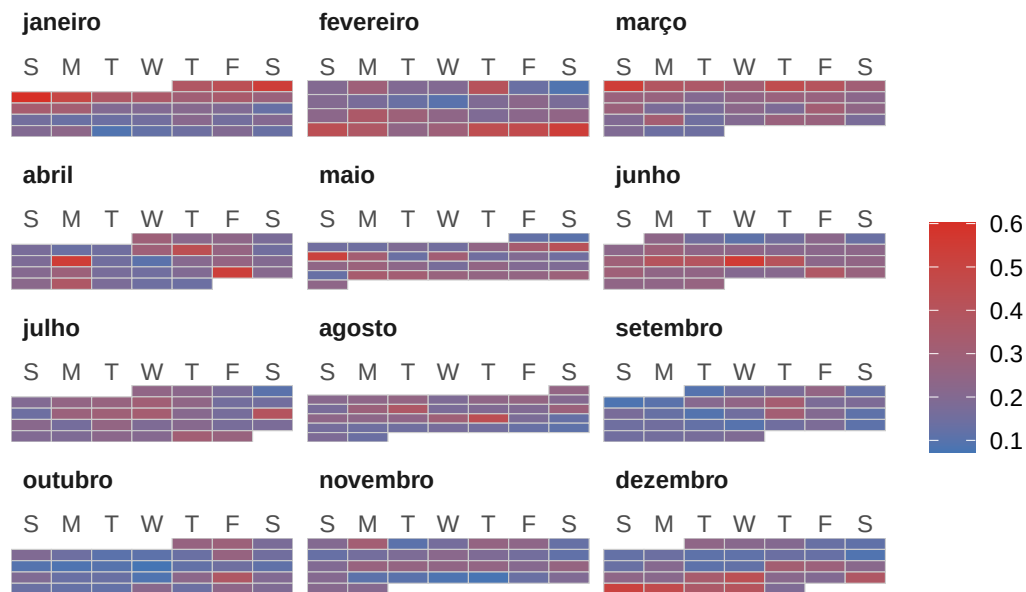
baseCalendario <- function(stats, cia) {
  dados_cia <- stats %>% filter(Cia == cia)
  ggcal(dados_cia$Data, dados_cia$Perc)
}

cAA <- baseCalendario(stats, "AA")
cDL <- baseCalendario(stats, "DL")
cUA <- baseCalendario(stats, "UA")
cUS <- baseCalendario(stats, "US")

cAA + pal + ggtitle("% de Atrasos - American Airlines (AA)")

```

% de Atrasos – American Airlines (AA)

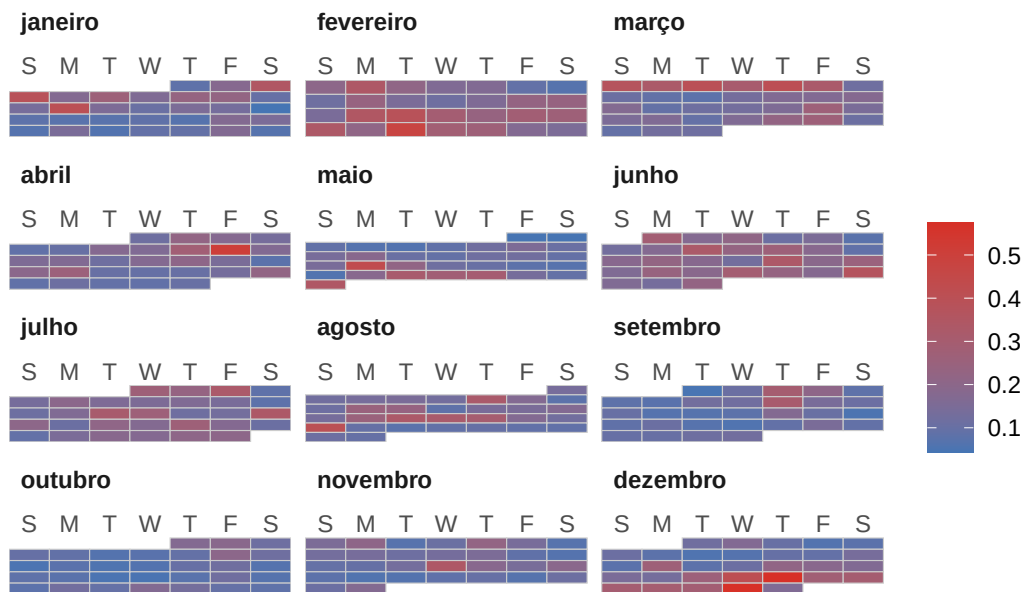


```

cDL + pal + ggtitle("% de Atrasos - Delta Air Lines (DL)")

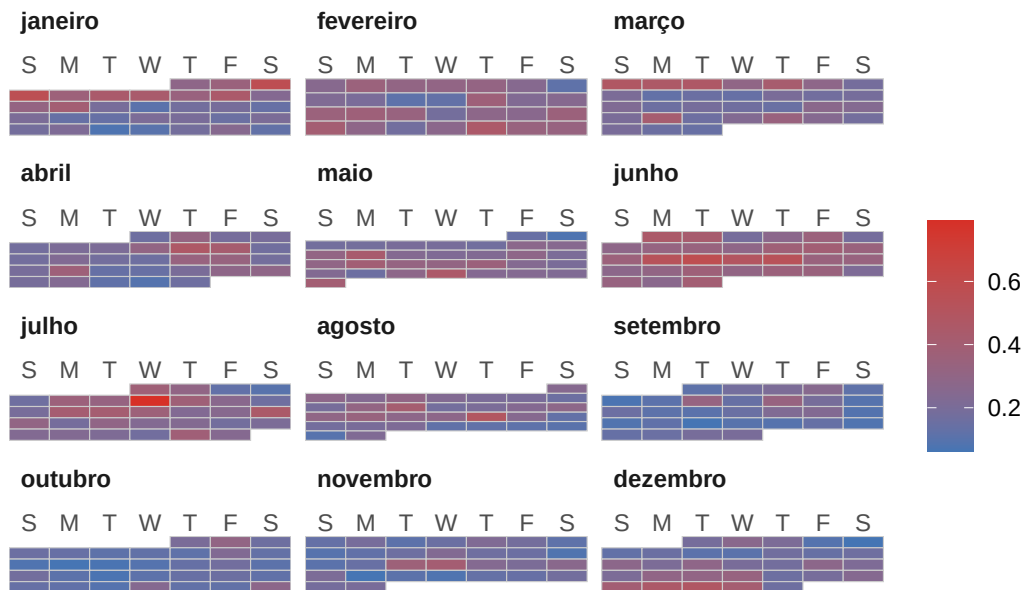
```

% de Atrasos – Delta Air Lines (DL)



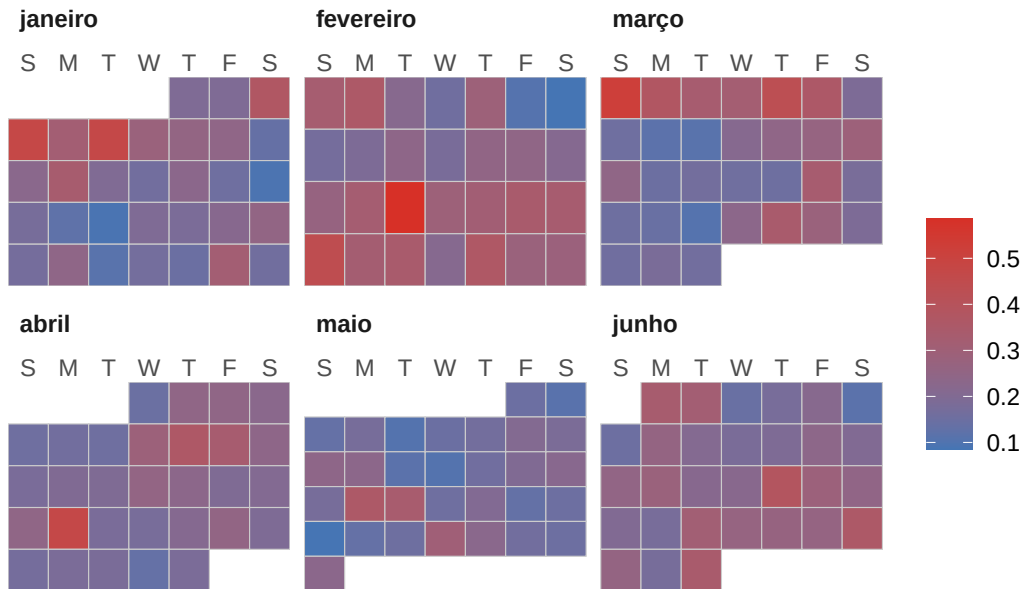
```
cUA + pal + ggtitle("% de Atrasos - United Airlines (UA)")
```

% de Atrasos – United Airlines (UA)



```
cUS + pal + ggtitle("% de Atrasos - US Airways (US)")
```

% de Atrasos - US Airways (US)



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-24 15:15:00 -03"
```